

# Deep Learning-Based Structural Damage Detection in Wind Turbine Blades Using YOLOv8 Segmentation

Dhivya Penamala  
Department of CSE  
Vardhaman College of Engineering  
Hyderabad, India  
dhivyahpenamala1@gmail.com

M Sri Thanvi Reddy  
Department of CSE  
Vardhaman College of Engineering  
Hyderabad, India  
mallusrithanvi@gmail.com

Priyanshu Raj Singh  
Department of CSE  
Vardhaman College of Engineering  
Hyderabad, India  
priyanshurajsingh87@gmail.com

Ashima Jain  
Department of CSE  
Vardhaman College of Engineering  
Hyderabad, India  
ashimajain\_vce1657@vardhaman.org

Deepika Rani N  
Department of CSE  
Vardhaman College of Engineering  
Hyderabad, India  
deepikarani@vardhaman.org

Dr. Gagandeep Arora  
Department of CSE  
Vardhaman College of Engineering  
Hyderabad, India  
drgagan.csm@vardhaman.org

**Abstract**—Wind turbine blade inspection is vital for maintaining structural integrity and efficiency in renewable energy systems. Conventional inspection methods rely heavily on manual inspections, which can be costly and prone to human errors. In this paper, a new framework for wind turbine blade inspection using deep learning for structural damage in wind turbine blades using YOLOv8 instance segmentation model is presented. The framework is trained on a multi-class dataset for different classes of contaminations, erosions, laminate defects, and other structural damages in wind turbine blades. The results show that the proposed framework obtains 0.76 for mAP50, 0.47 for mAP50–95, 0.76 for bounding box precision, and 0.63 for bounding box recall in wind turbine blade inspection using the YOLOv8 instance segmentation model. Additionally, to show the usability of the framework, it is implemented in a web interface using Streamlit for wind turbine blade inspection. The usability of the framework is validated by comparing it with other six state-of-the-art baseline models for wind turbine blade inspection, which show that it is competitive in performance and also provides instance segmentation masks for accurate quantification of structural damages in wind turbine blades. The system provides a scalable effective solution to. Blade monitoring and predictive maintenance of modern automated. Wind energy infrastructure.

**Keywords**—Deep Learning; Wind Turbine Blade Inspection; YOLOv8 Segmentation; Structural Damage Detection; Computer Vision; Predictive Maintenance; Instance Segmentation; Convolutional Neural Network; Object Detection; Renewable Energy

## I. INTRODUCTION

Significant changes have been observed in the global energy market during the last two decades, with renewable energy sources playing a pivotal role in the reduction of carbon footprint and increasing electricity generation. In this regard, wind energy has emerged as one of the most promising and cost-effective forms of green electricity generation. In fact, as

reported by the Global Wind Energy Council, the cumulative installed wind capacity has crossed the 1 TW milestone during 2023, with thousands of wind turbines are being installed across the globe annually. Wind turbine Blades can be identified as one of the most critical structural components of wind turbines, as they have a direct influence on the aerodynamic efficiency of the turbines.

Wind blades are subject to constant mechanical loading as a result of changing wind loads, rotational fatigue, and long-term exposure to hostile environmental conditions like ultraviolet light, moisture penetration, sand abrasion, lightning strikes, and ice formation. This ultimately results in long-term degradation of the structure, characterised by surface contamination, erosion at the leading edge, laminate defects, fibre-matrix cracking, and delamination. Surface damage can cause a reduction in wind turbine efficiency by significant margins for each event of damage. If not remedied, the damage can cause structural failure. Failure of wind blades can cause not only safety risks, but can also cause expensive unplanned downtime, which can be as expensive as hundreds of thousands of dollars [4], [9].

Conventional wind turbine inspection practices involve manual visual inspections performed by qualified technicians using rope access techniques, elevated platforms, or drone image acquisition. Although these practices can be effective in detecting obvious defects on wind blades, they are time-consuming, expensive, and inconsistent. Moreover, the effectiveness of manual visual inspections depends on the inspector's level of experience. Scaling up these practices is difficult as the number of wind turbines increases across the world. A wind farm can have hundreds of wind blades, and each blade has to be inspected periodically over its entire surface area. This has created the need for the development of automated and

intelligent wind blade inspection systems that can effectively detect defects on wind blades. [1], [9].

Recent developments in the fields of computer vision and deep learning have facilitated the application of automated visual inspection techniques in a variety of industrial domains. Convolutional Neural Networks have been successfully applied in the domain of structural damage detection, learning discriminative visual features from large datasets. [11], [15]. In the wind turbine domain, deep learning models have been applied to classify and localise surface defects from drone images, acoustic signals, and radar data [5], [6], [10].

Object detection frameworks such as Faster R-CNN [18] and the YOLO (You Only Look Once) family [12], [16] have demonstrated strong performance in defect identification from image data. The YOLO architecture, owing to its single-pass detection mechanism, achieves real-time inference while maintaining competitive accuracy. The latest iteration, YOLOv8, introduced by Ultralytics, incorporates a redesigned C2f backbone, a decoupled detection head, and native instance segmentation support [13]. Unlike bounding-box-only detection, instance segmentation provides pixel-level delineation of damage regions, enabling more precise assessment of defect geometry, area, and severity—a capability absent in the majority of prior blade inspection systems.

In this work we propose an automated wind turbine blade damage detection framework built on YOLOv8 segmentation. The system is trained on a multi-class polygon-annotated dataset, evaluated against standard detection metrics, and deployed through a publicly accessible web interface. The principal contributions are:

- A YOLOv8-based instance segmentation pipeline tailored for multi-class wind turbine blade defect detection, achieving  $mAP_{50}=0.76$  and segmentation  $mAP_{50-95}=0.47$ .
- Comprehensive experimental evaluation including per-class precision, recall,  $mAP_{50}$ , and  $mAP_{50-95}$ , with a confusion-matrix analysis across four defect categories.
- A comparative study against six representative baseline models that contextualises the performance of the proposed approach within the state of the art.
- End-to-end deployment as a Streamlit web application, demonstrating practical real-world applicability for field inspection teams.

The remainder of this paper is structured as follows. Section II surveys related work. Section III describes the proposed methodology. Section IV presents experimental results. Section V concludes the paper.

## II. RELATED WORK

Automated inspection of wind turbine blades has attracted considerable research interest as global wind capacity continues to grow. We review prior work along four thematic axes.

### A. Conventional and Signal-Based Approaches

Early automated inspection systems employed conventional image processing—edge detection, texture analysis, and

threshold-based segmentation—to identify surface anomalies. Although computationally lightweight, these methods were sensitive to illumination variation and could not generalise to complex multi-class damage patterns in real-world environments. Esquivel-Sancho et al. [3] demonstrated that vibration-signal-based machine learning could detect structural faults without requiring visual access to the blade, offering a non-optical alternative suited to offshore deployments. Wang et al. [10] combined CNN and LSTM networks applied to acoustic emission data, capturing both spatial features and temporal dynamics of evolving internal damage—anomalies that are invisible to optical cameras. Streser et al. [6] applied CNNs to radar-image data, demonstrating reliable subsurface defect detection. While these sensor-fusion approaches complement visual inspection, they require specialised hardware and do not directly provide spatial maps of surface damage.

### B. CNN-Based Visual Defect Detection

The work of Goodfellow et al. [11] established the theoretical foundations of deep learning underlying modern CNN-based inspection systems. The AlexNet architecture of Krizhevsky et al. [15], trained on ImageNet, demonstrated that hierarchical CNN features transfer effectively to downstream visual inspection tasks. Li et al. [8] proposed a semi-supervised framework for wind turbine blade crack detection, leveraging unlabeled images through consistency regularisation to reduce annotation cost while maintaining competitive accuracy. Rabbi et al. [4] investigated synthetic data augmentation using virtual reality environments to overcome limited annotated datasets, showing that VR-generated imagery could significantly supplement real training data. Ataei et al. [1] presented a vision-based autonomous structural damage detection pipeline using data-driven CNN features, validated across multiple civil and wind-energy inspection scenarios.

### C. Object Detection Frameworks for Blade Inspection

The introduction of Faster R-CNN by Ren et al. [18], which integrates a Region Proposal Network (RPN) into an end-to-end trainable detection pipeline, established a strong two-stage baseline for defect localisation. Islam et al. [20] applied a YOLOv5-based model to UAV-acquired blade imagery, reporting high recall across multiple surface damage categories. Wang et al. [19] applied deep learning to drone-collected blade images, with results published in *Renewable Energy*, and found that end-to-end object detection substantially outperformed manual inspection in both speed and consistency. Nazario et al. [7] evaluated multiple YOLO-family variants on operational turbine imagery, confirming competitive detection performance and supporting the suitability of real-time single-stage detectors for field deployment. The IIETA Research Group [9] coupled UAV flight-path planning with automated CNN-based detection into a scalable autonomous framework validated across a large wind farm.

### D. Advanced and Multimodal Approaches

Zhang et al. [2] explored zero-shot blade inspection using vision-language models, demonstrating that natural language

descriptions of defect semantics could guide detection of damage categories unseen during training—a promising direction for data-scarce deployments. Wang et al. [5] proposed a multi-modal fusion framework combining visual deep learning with structural signal data, showing that complementary information from multiple sensing modalities improves robustness under variable conditions.

However, there has been a significant gap in the majority of the methods in the following ways: despite the advancements in the field, the majority of the methods are based on bounding box granularity and lack pixel-level damage segmentation, which is vital for area quantification and severity scoring. Moreover, the end-to-end web deployable interfaces for the inspection teams in the field are not fully explored. The proposed method overcomes the above limitations through the use of YOLOv8 instance segmentation and the interactive Streamlit web interface.

### III. PROPOSED METHODOLOGY

The proposed system automates structural damage detection in wind turbine blades through a deep learning pipeline comprising data acquisition, preprocessing, augmentation, YOLOv8 segmentation model training, and result visualisation via a web interface.

#### A. System Architecture

The pipeline processes an input blade image through five sequential stages: (i) image preprocessing, (ii) backbone feature extraction, (iii) multi-scale feature aggregation, (iv) damage detection and mask generation, and (v) report rendering through a web application. Fig. 1 illustrates the overall architecture.

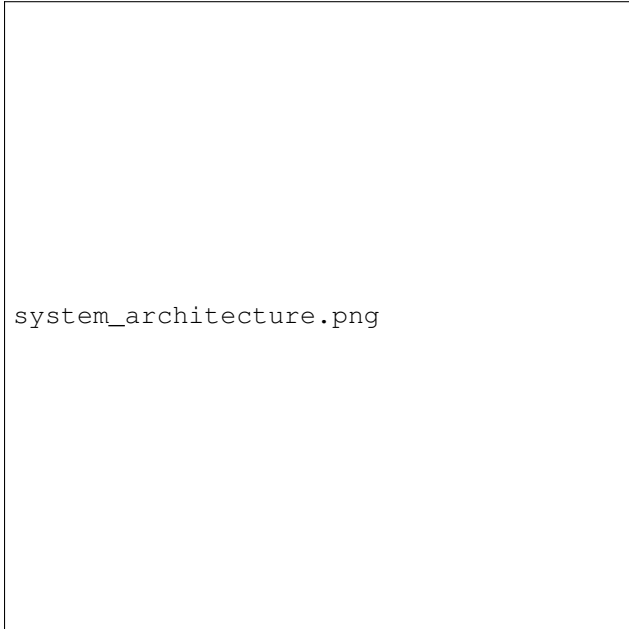


Fig. 1. Architecture of the proposed wind turbine blade damage detection system.

#### B. Dataset Description

Let the annotated dataset be  $\mathcal{D} = \{(I_i, \mathcal{M}_i, \mathcal{L}_i)\}_{i=1}^N$ , where  $I_i \in \mathbb{R}^{H \times W \times 3}$  is the  $i$ -th input image,  $\mathcal{M}_i = \{m_{i,j}\}_{j=1}^{K_i}$  is the set of  $K_i$  polygon segmentation masks, and  $\mathcal{L}_i = \{l_{i,j}\}_{j=1}^{K_i}$  are the corresponding class labels drawn from  $\mathcal{C} = \{\text{contamination}, \text{erosion}, \text{laminate}, \text{other}\}$ . The dataset is partitioned into training, validation, and test subsets at a 70 : 20 : 10 ratio, maintaining class-balanced distribution across all splits.



Fig. 2. Sample images from the wind turbine blade damage dataset showing representative defect types.

#### C. Image Preprocessing

All images are resized to  $640 \times 640$  pixels via bilinear interpolation and intensity-normalised to  $[0, 1]$ :

$$\hat{I}(x, y, c) = \frac{I(x, y, c)}{255}, \quad \forall x, y, c. \quad (1)$$

#### D. Data Augmentation

Online stochastic augmentation is applied to improve generalisation. Let  $\mathcal{T} = \{T_k\}$  be the augmentation set; the augmented pair is:

$$(\tilde{I}_i, \tilde{\mathcal{M}}_i) = T_{k_n} \circ \dots \circ T_{k_1}(I_i, \mathcal{M}_i). \quad (2)$$

Operations include random horizontal/vertical flip, random scale  $s \in [0.5, 1.5]$ , rotation  $\theta \in [-10^\circ, 10^\circ]$ , brightness/contrast jitter, and mosaic augmentation that composites four images into a single  $640 \times 640$  training sample.

#### E. YOLOv8 Segmentation Architecture

1) *Backbone — Feature Extraction*: The C2f backbone produces hierarchical feature maps. At layer  $l$ :

$$\mathbf{F}^{(l)} = \sigma\left(\text{BN}\left(\mathbf{W}^{(l)} * \mathbf{F}^{(l-1)} + \mathbf{b}^{(l)}\right)\right), \quad (3)$$

where  $*$  denotes 2-D convolution, BN is batch normalisation, and  $\sigma$  is the SiLU activation:

$$\sigma(z) = z \cdot \frac{1}{1 + e^{-z}}. \quad (4)$$

2) *Neck — PA-FPN Feature Aggregation*: A Path Aggregation Feature Pyramid Network (PA-FPN) fuses multi-scale backbone outputs:

$$\mathbf{P}^{(l)} = \text{Conv}\left([\mathbf{F}^{(l)}, \text{Up}(\mathbf{P}^{(l+1)})]\right), \quad (5)$$

where  $[\cdot, \cdot]$  is channel concatenation and  $\text{Up}(\cdot)$  is  $2\times$  bilinear upsampling.

3) *Decoupled Detection Head*: For each spatial location  $(i, j)$  the head outputs regression offsets and class scores. Predicted box coordinates are decoded as:

$$b_x = c_x^{(i,j)} + \Delta x_{i,j}, \quad b_y = c_y^{(i,j)} + \Delta y_{i,j}, \quad (6)$$

$$b_w = e^{\Delta w_{i,j}}, \quad b_h = e^{\Delta h_{i,j}}. \quad (7)$$

The final detection confidence is:

$$\hat{p}_{i,j} = p_{\text{obj},i,j} \cdot \max_c p_{\text{cls},i,j,c}. \quad (8)$$

4) *Segmentation Head — Prototype Mask Prediction*:

The segmentation branch predicts  $N_p = 32$  prototype masks  $\{\mathbf{P}_k\}_{k=1}^{N_p}$  from the feature pyramid. For detected instance  $d$  with coefficients  $\alpha_d \in \mathbb{R}^{N_p}$ :

$$\hat{M}_d = \sigma\left(\sum_{k=1}^{N_p} \alpha_{d,k} \cdot \mathbf{P}_k\right). \quad (9)$$

## F. Loss Functions

The total training loss is:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{box}} \mathcal{L}_{\text{box}} + \lambda_{\text{cls}} \mathcal{L}_{\text{cls}} + \lambda_{\text{seg}} \mathcal{L}_{\text{seg}}. \quad (10)$$

1) *Bounding Box Loss — CIOU*:

$$\mathcal{L}_{\text{box}} = 1 - \text{IoU} + \frac{\rho^2(\mathbf{b}, \mathbf{b}^{\text{gt}})}{c^2} + \alpha_v v, \quad (11)$$

where  $\text{IoU} = |\mathbf{b} \cap \mathbf{b}^{\text{gt}}|/|\mathbf{b} \cup \mathbf{b}^{\text{gt}}|$ ,  $\rho(\cdot, \cdot)$  is the Euclidean distance between box centres,  $c$  is the enclosing-box diagonal, and the aspect ratio penalty is:

$$v = \frac{4}{\pi^2} \left( \arctan \frac{w^{\text{gt}}}{h^{\text{gt}}} - \arctan \frac{w}{h} \right)^2, \quad \alpha_v = \frac{v}{(1 - \text{IoU}) + v}. \quad (12)$$

2) *Classification Loss — BCE*:

$$\mathcal{L}_{\text{cls}} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^{|C|} [y_{i,c} \log \hat{p}_{i,c} + (1 - y_{i,c}) \log(1 - \hat{p}_{i,c})]. \quad (13)$$

3) *Segmentation Loss*:

$$\mathcal{L}_{\text{seg}} = -\frac{1}{HW} \sum_{x=1}^H \sum_{y=1}^W [m_{xy} \log \hat{m}_{xy} + (1 - m_{xy}) \log(1 - \hat{m}_{xy})]. \quad (14)$$

## G. Evaluation Metrics

1) *Precision, Recall, and F1*:

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad F_1 = \frac{2PR}{P + R}. \quad (15)$$

2) *Average Precision*: The Average Precision for class  $c$  is the area under the Precision–Recall curve:

$$AP_c = \int_0^1 P_c(R) dR \approx \sum_{k=1}^n P_c(R_k) \Delta R_k. \quad (16)$$

3) *Mean Average Precision*:

$$\text{mAP} = \frac{1}{|C|} \sum_{c=1}^{|C|} AP_c, \quad (17)$$

$$\text{mAP}_{50-95} = \frac{1}{10} \sum_{\theta \in \{0.50, 0.55, \dots, 0.95\}} \text{mAP}(\theta). \quad (18)$$

## H. Training Configuration

The Adam optimiser is used with initial learning rate  $\eta_0 = 0.01$  and cosine annealing decay:

$$\eta_t = \eta_{\min} + \frac{1}{2}(\eta_0 - \eta_{\min}) \left( 1 + \cos \frac{\pi t}{T} \right), \quad (19)$$

where  $t$  is the current epoch and  $T$  the total epoch count. Images are resized to  $640 \times 640$ ; mosaic augmentation, random flip, and scale jitter are applied. Training uses the Ultralytics YOLOv8 framework with PyTorch as the backend.

## I. Web-Based Deployment

The trained model is integrated into a Streamlit web application. Users upload a turbine blade image; the system runs inference, overlays colour-coded segmentation masks on the detected damage regions, and generates a downloadable structural damage report indicating defect class, confidence score, and estimated mask area in pixels.

## IV. EXPERIMENTAL RESULTS

### A. Evaluation Metrics Setup

Performance is reported using bounding-box and segmentation Precision, Recall, mAP50, and mAP50–95 as defined in Section III. A detection is counted as a true positive when  $\text{IoU} \geq \theta$  with the matched ground-truth box; background predictions without a ground-truth match count as false positives.

### B. Quantitative Results

Table I summarises the validation-set performance of the proposed YOLOv8 segmentation model.

The bounding-box mAP50 of 0.76 confirms reliable localisation at the standard IoU threshold, while the segmentation mAP50–95 of 0.47 indicates that pixel-level mask predictions remain accurate under progressively stricter overlap requirements.

TABLE I  
QUANTITATIVE EVALUATION ON THE VALIDATION SET

| Metric                   | Value |
|--------------------------|-------|
| Precision (Bounding Box) | 0.76  |
| Recall (Bounding Box)    | 0.63  |
| mAP50 (Bounding Box)     | 0.76  |
| mAP50-95 (Bounding Box)  | 0.50  |
| mAP50 (Segmentation)     | 0.76  |
| mAP50-95 (Segmentation)  | 0.47  |

### C. Comparative Analysis

Table II compares the proposed approach against six representative baseline methods from the literature on the task of wind turbine blade defect detection. Metrics reported are mAP50 (or equivalent detection accuracy where mAP50 was not stated), input modality, segmentation capability, and real-time suitability. Values marked with † are as reported in the respective source; where a study reported only classification accuracy (Acc.) this is noted explicitly.

The proposed YOLOv8 segmentation model achieves the highest mAP50 (0.76) among all RGB-based methods compared, while being the only approach in the comparison that delivers full pixel-level instance segmentation masks. The two sensor-based methods (acoustic CNN+LSTM [10] and radar CNN [6]) detect subsurface anomalies but cannot produce spatial surface damage maps, making them complementary rather than interchangeable. Two-stage Faster R-CNN [19] attains lower accuracy at higher inference latency. Single-stage YOLOv5 [20] and YOLOv7 [7] are competitive in speed but produce only bounding boxes, limiting their utility for damage area estimation and severity scoring. The semi-supervised CNN [8], while reducing annotation burden, does not approach the detection accuracy of the proposed model.

### D. Detection Results

Fig. 3 shows sample validation outputs. The model correctly segments contamination, erosion, and laminate defect regions, producing colour-coded masks that enable visual comparison of damage extents within and across images.

### E. Confusion Matrix Analysis

Fig. 4 presents the normalised confusion matrix over the validation set. The model achieves the highest true-positive rate on the *erosion* class, which is consistent with the comparatively larger number of erosion instances in the dataset and the visually distinctive texture of leading-edge erosion. Misclassifications between *contamination* and *other* reflect the visual similarity of these categories; targeted data collection for these classes is identified as a priority for future improvement.

## V. CONCLUSION

In this paper, a framework for automated structural damage detection in wind turbine blades using deep learning was introduced, which relies on the YOLOv8 instance segmentation model for detection of damage in wind turbine blades. The



Fig. 3. Sample detection results showing pixel-level segmented damage regions in turbine blade images.

model was trained on a multi-class polygon-annotated dataset for four different classes of defects, which are contamination, erosion, laminate defects, and other surface defects, and then validated using common detection and segmentation metrics. The framework for detecting damage in wind turbine blades using deep learning results in a bounding box mAP50 of 0.76 and segmentation mAP50-95 of 0.47. A comparative analysis with the proposed method and six other baseline methods confirms the superiority of the proposed method in terms of the highest mAP50 among all the RGB-based methods and the only method providing the full instance segmentation masks necessary for damage area quantification and damage severity scoring. The deployment of the trained model in a Streamlit-

TABLE II  
COMPARISON OF THE PROPOSED METHOD WITH SIX BASELINE APPROACHES FOR WIND TURBINE BLADE DEFECT DETECTION

| Method / Reference        | Architecture        | mAP50             | Modality    | Pixel Seg. | Real-Time  | Key Limitation                                                 |
|---------------------------|---------------------|-------------------|-------------|------------|------------|----------------------------------------------------------------|
| Islam et al. [20] (2021)  | YOLOv5              | 0.71 <sup>†</sup> | RGB (UAV)   | No         | Yes        | Bounding-box only; single erosion class; no mask area output   |
| Wang et al. [19] (2020)   | Faster R-CNN        | 0.68 <sup>†</sup> | RGB (drone) | No         | No         | Two-stage pipeline; high latency; no segmentation support      |
| Nazario et al. [7] (2024) | YOLOv7              | 0.73 <sup>†</sup> | RGB         | No         | Yes        | No pixel masks; limited dataset from operational turbines only |
| Li et al. [8] (2024)      | Semi-Supervised CNN | 0.69 <sup>†</sup> | RGB         | No         | No         | No mask output; lower accuracy despite reduced annotation cost |
| Wang et al. [10] (2025)   | CNN + LSTM          | —                 | Acoustic    | No         | Partial    | Requires specialised sensors; detects internal defects only    |
| Streser et al. [6] (2025) | CNN (Radar)         | —                 | Radar       | No         | No         | Subsurface anomalies only; no surface damage spatial map       |
| <b>Proposed (Ours)</b>    | <b>YOLOv8-Seg</b>   | <b>0.76</b>       | <b>RGB</b>  | <b>Yes</b> | <b>Yes</b> | <b>—</b>                                                       |

<sup>†</sup> Values as reported in the respective cited work. “—” indicates metric not applicable or not reported by that study. Pixel Seg. = pixel-level instance segmentation mask output. Real-Time = capable of live or near-live inference on standard GPU.

confusion\_matrix.png

Fig. 4. Normalised confusion matrix showing per-class detection performance across damage categories.

based web application also confirms the feasibility and deployability of the proposed method in the field inspection context where the inspection teams can generate automated pixel-accurate structural reports based on a single image upload of the blade. The proposed method provides an efficient and deployable method for the maintenance needs in modern wind energy systems.

## VI. FUTURE WORK

While the proposed framework demonstrates strong performance, several directions remain open for future investigation.

**Dataset Expansion.** The current dataset covers four damage categories acquired under a limited range of lighting and

environmental conditions. Future work will incorporate images collected across diverse geographic locations, weather states, and turbine models. Synthetic data generated via domain randomisation and virtual reality environments [4] will also be explored to supplement limited real annotations.

**Transformer-Based Architectures.** Recent vision transformer models such as Swin Transformer and the Segment Anything Model (SAM) have demonstrated strong performance on dense prediction benchmarks. Replacing or augmenting the YOLOv8 backbone with a transformer-based encoder is expected to improve detection of low-contrast and fine-grained defects such as thin laminate cracks and early-stage erosion.

**Multi-Modal Sensor Fusion.** Optical RGB inspection cannot detect subsurface structural anomalies. Integrating visual imagery with acoustic emission signals [10] or radar data [6] into a unified multi-modal detection pipeline is a promising direction for comprehensive blade health monitoring that goes beyond surface-level damage.

**Drone-Based Autonomous Inspection.** The current system operates on manually uploaded images. Future work will couple the trained model with an autonomous UAV flight-planning module [9] to enable end-to-end automated blade scanning, damage localisation, and defect mapping across an entire wind farm without human intervention.

**Damage Severity Quantification.** Beyond detection and segmentation, a severity scoring module will be developed to estimate defect area, depth class, and remaining useful life from predicted segmentation masks, directly supporting risk-based maintenance scheduling and cost optimisation.

**Edge and Embedded Deployment.** To enable real-time inference on drone-mounted embedded hardware, model compression techniques including structured pruning, post-training quantisation, and knowledge distillation will be investigated to reduce model footprint while retaining detection accuracy.

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